

Verification Plan Report

Project: AXI-Stream Master VIP Verification Plan (Advanced Iteration)

Version: 1.1

Verification Role: Master

1. UVM Environment Architecture

```
graph TD
    classDef test_layer fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px,color:#7b1fa2;
    classDef test_obj fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px,stroke-dasharray: 5
5,color:#7b1fa2;
    classDef env_layer fill:#e3f2fd,stroke:#1565c0,stroke-width:2px,color:#1565c0;
    classDef env_obj fill:#e3f2fd,stroke:#1565c0,stroke-width:2px,stroke-dasharray: 5
5,color:#1565c0;
    classDef agent_layer fill:#e0f2f1,stroke:#00695c,stroke-width:2px,color:#00695c;
    classDef agent_obj fill:#e0f2f1,stroke:#00695c,stroke-width:2px,stroke-dasharray: 5
5,color:#00695c;
    classDef dut_layer fill:#fff8e1,stroke:#f57f17,stroke-width:2px,color:#f57f17;

    subgraph UVM_Test_Layer [axi_stream_test]
        SeqItem[axi_stream_seq_item]:::test_obj
        Seq[axi_stream_master_sequence]:::test_obj
        subgraph Env_Layer [axi_stream_env]
            Scoreboard[axi_stream_scoreboard]:::env_layer
            Env_Cfg[axi_stream_env_config]:::env_obj
            subgraph Agent_Layer [Master Agent (VIP)]
                Sequencer[axi_stream_master_sequencer]:::agent_layer
                Agent_Cfg[axi_stream_master_agent_config]:::agent_obj
                Driver[axi_stream_master_driver]:::agent_layer
                Monitor[axi_stream_master_monitor]:::agent_layer
                Sequencer --> Driver
                Agent_Cfg --> Sequencer
            end
        end
    end

    subgraph Top_Level [Hardware Top]
        VIF[Virtual Interface (axi_stream_if)]:::dut_layer
        DUT[DUT Slave Receiver]:::dut_layer
        VIF <--> DUT
    end

    Driver --- VIF
    Monitor --- VIF
    Monitor --> Scoreboard
```

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2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MST_01	Handshake Protocol (Reset Conformance)	Reset Exit Timing	TVALID must be deasserted during ARESETn LOW and must not be asserted until the first rising clock edge after ARESETn goes HIGH. Verification intent is to verify that the VIP does not start driving valid data prematurely upon reset release.
REQ_MST_02	Data Streams (Byte Qualifiers)	Data vs. Position Byte Handling	TKEEP determines if a byte persists to the destination, while TSTRB differentiates between valid data bytes and 'position bytes'. Verification strategy involves generating sparse transfers with interleaved TSTRB/TKEEP bits to verify correct Slave extraction logic.
REQ_MST_03	Streaming (Interleaving)	Per-Transfer Stream Interleaving	Master must be capable of switching between different streams (unique TID/TDEST) without waiting for a TLAST boundary. Verification intent is to toggle TID/TDEST values on consecutive TVALID assertions and verify the Slave/Interconnect's routing logic.
REQ_MST_04	Packet boundaries (Null Transfer)	Zero-Data Packet Handling	A transfer can have TLAST asserted with zero TKEEP bits asserted. This acts as a 'flush' or end-of-packet marker. Verification involves injecting these 'Null Transfers' and ensuring they don't break Slave synchronization.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	Reset Conformance: Assert reset during a burst. Verify TVALID drops and stays low for 1 cycle after reset exit.	REQ_MST_01
TC_MST_002	Sparse Stream: Send TDATA with random TSTRB/TKEEP masks. Verify Scoreboard handles position-byte exclusion.	REQ_MST_02
TC_MST_003	Interleaved Multi-ID: Interleave 4 different TIDs on consecutive cycles (ID 0, ID 1, ID 0, ID 2).	REQ_MST_03
TC_MST_004	Flush Packet: Send a packet finishing with a TLAST-asserted transfer where TKEEP=0.	REQ_MST_04

4. Functional Coverage Groups

Group Name	Description
cg_reset_exit	Cover points for TVALID state on first cycle after reset.
cg_byte_masks	Cross coverage of all TKEEP/TSTRB bit combinations.

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5. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input	1	Global clock.
ARESETn	Input	1	Active low reset.
TVALID	Output	1	Valid transfer indicator.
TREADY	Input	1	Slave ready indicator.
TDATA	Output	32	Payload.
TKEEP	Output	4	Byte validity qualifier.
TSTRB	Output	4	Position byte vs Data byte qualifier.
TID	Output	8	Stream identifier.
TLAST	Output	1	Packet boundary.

6. Verification Environment Structure

axi_stream_master_vip_v1.1/

```
|-- master_agent/  
|   |-- axi_stream_master_driver.sv  
|   |-- axi_stream_master_monitor.sv  
|   |-- axi_stream_master_sequencer.sv  
|-- env/  
|   |-- axi_stream_scoreboard.sv  
|   |-- axi_stream_env_config.sv  
|-- top/  
    |-- axi_stream_master_tb_top.sv  
    |-- axi_stream_master_test.sv
```

7. SystemVerilog Implementation

axi_stream_master_driver.sv (Reset Handling)

```
virtual task reset_phase(uvm_phase phase);  
    vif.TVALID <= 1'b0;  
    vif.TDATA  <= 'h0;  
    wait(vif.ARESETn == 1'b1);  
    @(posedge vif.ACLK); // Spec Requirement: Stay Low for 1 cycle  
endtask
```